Chapter 5 : Molecular Basis of Inheritance

1. Which chromosome in the human genome has the highest number of genes? **[NEET-2025]**
   (1) Chromosome X
   (2) Chromosome Y
   (3) Chromosome 1
   (4) Chromosome 10

2. Match List I with List II: **[NEET-2025]**

List-I	List-II
A. Alfred Hershey and Martha Chase	I. Streptococcus pneumoniae
B. Euchromatin	II. Densely packed and dark-stained
C. Frederick Griffith	III. Loosely packed and light-stained
D. Heterochromatin	IV. DNA as genetic material confirmation

   (1) A-II, B-IV, C-I, D-III
   (2) A-IV, B-II, C-I, D-III
   (3) A-IV, B-III, C-I, D-II
   (4) A-III, B-II, C-IV, D-I

3. Which of the following are the post-transcriptional events in an eukaryotic cell? **[NEET-2025]**
   A. Transport of pre-mRNA to cytoplasm prior to splicing.
   B. Removal of introns and joining of exons.
   C. Addition of methyl group at 5′ end of hnRNA.
   D. Addition of adenine residues at 3' end of hnRNA.
   E. Base pairing of two complementary RNAs.
   Choose the correct answer from the options given below :
   (1) A, B, C only
   (2) B, C, D only
   (3) B, C, E only
   (4) C, D, E only

4. Given below are two statements : **[NEET-2025]**

Statement I : In the RNA world, RNA is considered the first genetic material evolved to carry out essential life processes. RNA acts as a genetic material and also as a catalyst for some important biochemical reactions in living systems. Being reactive, RNA is unstable.

Statement II : DNA evolved from RNA and is a more stable genetic material. Its double helical strands being complementary, resist changes by evolving repairing mechanism.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

5. Given below are two statements : **[NEET-2025]**

Statement I : Transfer RNAs and ribosomal RNA do not interact with mRNA.

Statement II : RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

6. Who proposed that the genetic code for amino acids should be made up of three nucleotides?

(1) George Gamow (2) Francis Crick **[NEET-2025]**

(3) Jacque Monod (4) Franklin Stahl

7. Which factor is important for termination of transcription? **[NEET-2025]**

(1) α (alpha) (2) σ (sigma)

(3) ρ (rho) (4) γ (gamma)

8. In a chromosome, there is a specific DNA sequence, responsible for initiating replication. It is called as: **[Re-NEET-2024]**

(1) Recognition sequence

(2) Cloning site

(3) Restriction site

(4) ori site

9. Given below are two statements regarding RNA polymerase in prokaryotes. **[Re-NEET-2024]**

Statement I : In prokaryotes, RNA polymerase is capable of catalysing the process of elongation during transcription.

Statement II : RNA polymerase associate transiently with 'Rho' factor to initiate transcription.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

10. Given below are two statements: **[Re-NEET-2024]**

Statement I: In eukaryotes there are three RNA polymerases in the nucleus in addition to the RNA polymerase found in the organelles.

Statement II: All the three RNA polymerases in eukaryotic nucleus have different roles.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is correct but Statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are correct

(4) Both Statement I and Statement II are incorrect

11. Match List-I with List-II: **[Re-NEET-2024]**

List-I	List-II
A. Histones	I. Loosely packed chromatin
B. Nucleosome	II. Densely packed Chromatin
C. Euchromatin	III. Positively charged basic proteins
D. Heterochromatin	IV. DNA wrapped around histone octamer

Choose the correct answer from the options given below:

(1) A-IV, B-III, C-II, D-I (2) A-III, B-I, C-IV, D-II

(3) A-II, B-III, C-IV, D-I (4) A-III, B-IV, C-I, D-II

12. Given below are two statements : **[Re-NEET-2024]**

Statement I: In the lac operon, the z gene codes for beta-galactosidase which is primarily responsible for the hydrolysis of lactose into galactose and glucose.

Statement II: In addition to lactose, glucose or galactose can also induce lac operon.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is true but Statement II is false (2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true (4) Both Statement I and Statement II are false

13. A transcription unit in DNA is defined primarily by the three regions in DNA and these are with respect to upstream and down stream end; **[NEET-2024]**

(1) Repressor, Operator gene, Structural gene

(2) Structural gene, Transposons, Operator gene

(3) Inducer, Repressor, Structural gene

(4) Promotor, Structural gene, Terminator

14. The lactose present in the growth medium of bacteria is transported to the cell by the action of

(1) Beta-galactosidase (2) Acetylase **[NEET-2024]**

(3) Permease (4) Polymerase

15. Match List I with List II **[NEET-2024]**

List I	List II
A. Frederick Griffith	I. Genetic code
B. Francois Jacob & Jacque Monod	II. Semi-conservative mode of DNA replication
C. Har Gobind Khorana	III. Transformation
D. Meselson & Stahl	IV. Lac operon

Choose the correct answer from the options given below:

(1) A-III, B-II, C-I, D-IV

(2) A-III, B-IV, C-I, D-II

(3) A-II, B-III, C-IV, D-I

(4) A-IV, B-I, C-II, D-III

16. Which of the following statement is correct regarding the process of replication in E.coli? **[NEET-2024]**

(1) The DNA dependent DNA polymerase catalyses polymerization in one direction that is $3' \rightarrow 5'$

(2) The DNA dependent RNA polymerase catalyses polymerization in one direction, that is $5' \rightarrow 3'$

(3) The DNA dependent DNA polymerase catalyses polymerization in $5' \rightarrow 3'$ as well as $3' \rightarrow 5'$ direction

(4) The DNA dependent DNA polymerase catalyses polymerization in $5' \rightarrow 3'$ direction

17. Which one is the correct product of DNA dependent RNA polymerase to the given template? **[NEET-2024]**

3'TACATGGCAAATATCCATTCA5'

(1) 5'AUGUACCGUUUAUAGGUAAGU3'

(2) 5'AUGUAAAGUUUAUAGGUAAGU3'

(3) 5'AUGUACCGUUUAUAGGGAAGU3'

(4) 5'ATGTACCGTTTATAGGTAAGT3'

18. Match List I with List II: **[NEET-2024]**

List I	List II
A. RNA polymerase III	I. snRNPs
B. Termination of transcription	II. Promotor
C. Splicing of Exons	III. Rho factor
D. TATA box	IV. SnRNAs, Trna

Choose the correct answer from the options given below :

(1) A-II, B-IV, C-I, D-III

(2) A-III, B-II, C-IV, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-IV, B-III, C-I, D-II

19. Name the component that binds to the operator region of an operon and prevents RNA polymerase from transcribing the operon. **[NEET-2023-MANIPUR]**

(1) Repressor protein (2) Inducer

(3) Promotor (4) Regulator protein

20. The last chromosome sequenced in Human Genome Project was : **[NEET-2023-MANIPUR]**

(1) Chromosome 22 (2) Chromosome 14

(3) Chromosome 6 (4) Chromosome 1

21. Given below are two statements : **[NEET-2023-MANIPUR]**

Statement I : The process of copying genetic information from one strand of the DNA into RNA is termed as transcription.

Statement II : A transcription unit in DNA is defined primarily by the three regions in the DNA, i.e., a promotor, the structural gene and a terminator.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

22. Which scientist conducted an experiment with ^{32}P and ^{35}S labelled phages for demonstrating that DNA is the genetic material? **[NEET-2023-MANIPUR]**

(1) F. Griffith

(2) O.T. Avery, C.M. MacLeod and M. McCarty

(3) James D. Watson and F.H.C. Crick

(4) A.D. Hershey and M.J. Chase

23. Given below are two statements: **[NEET-2023-MANIPUR]**

Statement I: RNA being unstable, mutate at a faster rate.

Statement II: RNA can directly code for synthesis of proteins, hence can easily express the characters.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is correct but Statement II is false

(4) Statement I is incorrect but Statement II is true

24. Which one of the following acts as an inducer for lac operon? **[NEET-2023-MANIPUR]**

(1) Glucose (2) Galactose

(3) Sucrose (4) Lactose

25. With reference to Hershey and Chase experiments. Select the correct statements.

(A) Viruses grown in the presence of radioactive phosphorus contained radioactive DNA.

(B) Viruses grown on radioactive sulphur contained radioactive proteins.

(C) Viruses grown on radioactive phosphorus contained radioactive protein.

(D) Viruses grown on radioactive sulphur contained radioactive DNA.

(E) Viruses grown on radioactive protein contained radioactive DNA.

Choose the most appropriate answer from the options given below : **[NEET-2023-MANIPUR]**

(1) (A) and (C) only (2) (B) and (D) only

(3) (D) and (E) only (4) (A) and (B) only

26. The salient features of genetic code are : **[NEET-2023-MANIPUR]**

(A) The code is palindromic

(B) UGA act as initiator codon

(C) The code is unambiguous and specific

(D) The code is nearly universal

Choose the most appropriate answer from the options given below :

(1) (A) and (B) only (2) (C) and (D) only

(3) (A) and (D) only (4) (B) and (C) only

27. Expressed Sequence Tags (ESTs) refers to **[NEET-2023]**

(1) All genes that are expressed as RNA.

(2) All genes that are expressed as proteins.

(3) All genes whether expressed or unexpressed.

(4) Certain important expressed genes.

28. What is the role of RNA polymerase III in the process of transcription in Eukaryotes?

(1) Transcription of rRNAs (28S, 18S and 5.8S) **[NEET-2023]**

(2) Transcription of tRNA, 5S rRNA and snRNA

(3) Transcription of precursor of mRNA

(4) Transcription of only snRNAs

29. Unequivocal proof that DNA is the genetic material was first proposed by **[NEET-2023]**

(1) Frederick Griffith (2) Alfred Hershey and Martha Chase

(3) Avery, Macleoid and McCarthy (4) Wilkins and Franklin

30. Given below are two statements: **[NEET-2023]**

Statement I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.

Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is correct but Statement II is false.

(4) Statement I is incorrect but Statement II is true.

31. Given below are two statements: **[NEET-2023]**

Statement I: RNA mutates at a faster rate.

Statement II: Viruses having RNA genome and shorter life span mutate and evolve faster.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

32. Read the following statements and choose the set of correct statements: - **[NEET-2022]**

(a) Euchromatin is loosely packed chromatin

(b) Heterochromatin is transcriptionally active

(c) Histone octomer is wrapped by negatively charged DNA in nucleosome

(d) Histones are rich in lysine and arginine

(e) A typical nucleosome contains 400 bp of DNA helix

Choose the correct answer from the options given below:

(1) (a), (c), (d) Only (2) (b), (e) Only

(3) (a), (c), (e) Only (4) (b), (d), (e) Only

33. DNA polymorphism forms the basis of:- **[NEET-2022]**

(1) DNA finger printing
(2) Both genetic mapping and DNA finger printing
(3) Translation
(4) Genetic mapping

34. The process of translation of mRNA to proteins begins as soon as :- **[NEET-2022]**

(1) The larger subunit of ribosome encounters mRNA
(2) Both the subunits join together to bind with mRNA
(3) The tRNA is activated and the larger subunit of ribosome encounters mRNA
(4) The small subunit of ribosome encounters mRNA

35. If a geneticist uses the blind approach for sequencing the whole genome of an organism, followed by assignment of function to different segments, the methodology adopted by him is called as:-

(1) Gene mapping
(2) Expressed sequence tags **[NEET-2022]**
(3) Bioinformatics
(4) Sequence annotation

36. In an E.coli strain i gene gets mutated and its product can not bind the inducer molecule. If growth medium is provided with lactose, what will be the outcome? **[NEET-2022]**

(1) z, y, a genes will be transcribed
(2) z, y, a genes will not be translated
(3) RNA polymerase will bind the promoter region
(4) Only z gene will get transcribed

37. If the length of DNA molecule is 1.1 metres, what will be the approximate number of base pairs ?

(1) 6.6×10^9 bp
(2) 3.3×10^6 bp **[NEET-2022]**
(3) 6.6×10^6 bp
(4) 3.3×10^9 bp

38. Ten E.coli cells with 15N - dsDNA are incubated in medium containing 14N nucleotide. After 60 minutes, how many E.coli cells will have DNA totally free from 15N ? **[Re-NEET-2022]**

(1) 40 cells
(2) 60 cells
(3) 80 cells
(4) 20 cells

39. Match List - I with List - II :- **[Re-NEET-2022]**

List - I	List - II
(a) In lac operon i gene codes for	(i) transacetylase
(b) In lac operon z gene codes for	(ii) permease
(c) In lac operon y gene codes for	(iii) b-galactosidase
(d) In lac operon a gene codes for	(iv) Repressor

Choose the correct answer from the options given below :

(1) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)
(2) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
(3) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)
(4) (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

40. Given below are two statements: - **[Re-NEET-2022]**

Statement I: DNA polymerases catalyse polymerisation only in one direction, that is 5' → 3'

Statement II : During replication of DNA, on one strand the replication is continuous while on other strand it is discontinuous.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

41. Match List-I with List-II :- **[Re-NEET-2022]**

List-I	List-II
(a) Bacteriophage	(i) 48502 base φ ×174 pairs
(b) Bacteriophage	(ii) 5386 lambda nucleotides
(c) Escherichia coli	(iii) 3.3×10^9 base pairs
(d) Haploid content	(iv) 4.6×10^6 base of human DNA pairs

Choose the correct answer from the options given below:

(1) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv) (2) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

(3) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii) (4) (a) - (i), (b) - (ii), (c) - (iv), (d) - (iii)

42. If DNA contained sulphur instead of phosphorus and proteins contained phosphorus instead of sulfur, what would have been the outcome of Hershey and Chase experiment? **[Re-NEET-2022]**

(1) No radioactive sulfur in bacterial cells

(2) Both radioactive sulfur and phosphorus in bacterial cells

(3) Radioactive sulfur in bacterial cells

(4) Radioactive phosphorus in bacterial Cells

43. In lac operon, z gene codes for :- **[Re-NEET-2022]**

(1) β-galactosidase (2) Permease

(3) Repressor (4) Transacetylase

44. Complete the flow chart on central dogma:- **[NEET-2021]**

$$\textbf{(a)}\circlearrowright \text{DNA} \xrightarrow{\textbf{(b)}} \text{mRNA} \xrightarrow{\textbf{(c)}} \textbf{(d)}$$

(1) (a)-Replication; (b)-Transcription; (c)-Transduction; (d)-Protein

(2) (a)-Translation; (b)-Replication; (c)-Transcription; (d)-Transduction

(3) (a)-Replication; (b)-Transcription; (c)-Translation; (d)-Protein

(4) (a)-Transduction; (b)-Translation; (c)-Replication; (d)-Protein

45. Identify the correct statement:- **[NEET-2021]**

(1) In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA.

(2) RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.

(3) The coding strand in a transcription unit is copied to an mRNA.

(4) Split gene arrangement is characteristic of prokaryotes.

46. What is the role of RNA polymerase III in the process of transcription in eukaryotes? **[NEET-2021]**

(1) Transcribes rRNAs (28S, 18S and 5.8S) (2) Transcribes tRNA, 5s rRNA and snRNA

(3) Transcribes precursor of mRNA (4) Transcribes only snRNAs

47. DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as :- **[NEET-2021]**

(1) Satellite DNA (2) Repetitive DNA

(3) Single nucleotides (4) Polymorphic DNA

48. Which is the "Only enzyme" that has "Capability" to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes? **[NEET-2021]**

(1) DNA dependent DNA polymerase (2) DNA dependent RNA polymerase

(3) DNA Ligase (4) DNase

49. Which of the following RNAs is not required for the synthesis of protein? **[NEET-2021]**

(1) mRNA (2) tRNA

(3) rRNA (4) siRNA

50. If Adenine makes 30% of the DNA molecule, what will be the percentage of Thymine, Guanine and Cytosine in it? **[NEET-2021]**

(1) T : 20; G : 30; C : 20 (2) T : 20; G : 20; C : 30

(3) T : 30; G : 20; C : 20 (4) T : 20; G : 25; C : 25

51. Which of the following statements is correct ? **[NEET-2020]**

(1) Adenine does not pair with thymine

(2) Adenine pairs with thymine through two H-bonds

(3) Adenine pairs with thymine through one H-bond

(4) Adenine pairs with thymine through three H-bonds

52. If the distance between two consecutive base pairs is 0.34 nm and the total number of base pairs of a DNA double helix in a typical mammalian cell is 6.6 × 109 bp, then the length of the DNA is approximately: - **[NEET-2020]**

(1) 2.7 meters (2) 2.0 meters

(3) 2.5 meters (4) 2.2 meters

53. Name the enzyme that facilitates opening of DNA helix during transcription: - **[NEET-2020]**

(1) RNA polymerase (2) DNA ligase

(3) DNA helicase (4) DNA polymerase

54. The first phase of translation is :- **[NEET-2020]**

(1) Recognition of an anti-codon (2) Binding of mRNA to ribosome

(3) Recognition of DNA molecule (4) Aminoacylation of tRNA

55. In gel electrophoresis, separated DNA fragments can be visualized with the help of **[NEET-2020]**

(1) Ethidium bromide in infrared radiation

(2) Acetocarmine in bright blue light

(3) Ethidium bromide in UV radiation

(4) Acetocarmine in UV radiation

56. The term 'Nuclein' for the genetic material was used by :- **[NEET - 2020 (COVID-19)]**

(1) Franklin (2) Meischer

(3) Chargaff (4) Mendel

57. In the polynucleotide chain of DNA, a nitrogenous base is linked to the –OH of:-

(1) 2'C pentose sugar (2) 3'C pentose sugar **[NEET - 2020 (COVID-19)]**

(3) 5'C pentose sugar (4) 1'C pentose sugar

58. E.coli has only 4.6×10^6 base pairs and completes the process of replication within 18 minutes; then the average rate of polymerisation is approximately: - **[NEET - 2020 (COVID-19)]**

(1) 2000 base pairs/second (2) 3000 base pairs/second

(3) 4000 base pairs/second (4) 1000 base pairs/second

59. Which is the basis of genetic mapping of human genome as well as DNA finger printing ?

(1) Polymorphism in DNA sequence **[NEET - 2020 (COVID-19)]**

(2) Single nucleotide polymorphism

(3) Polymorphism in hnRNA sequence

(4) Polymorphism in RNA sequence

60. Purines found both in DNA and RNA are :- **[NEET-2019]**

(1) Adenine and thymine (2) Adenine and guanine

(3) Guanine and cytosine (4) Cytosine and thymine

61. Under which of the following conditions will there be no change in the reading frame of following mRNA ? **[NEET-2019]**

5'AACAGCGGUGCUAUU3'

(1) Insertion of G at 5th position

(2) Deletion of G from 5th position

(3) Insertion of A and G and 4th and 5th positions respectively

(4) Deletion of GGU from 7th, 8th and 9th positions

62. Expressed Sequence Tags (ESTs) refers to :- **[NEET-2019]**

(1) Genes expressed as RNA (2) Polypeptide expression

(3) DNA polymorphism (4) Novel DNA sequences

63. Match the following genes of the Lac operon with their respective products :- **[NEET-2019]**

(a) i gene (i) β-galactosidase

(b) z gene (ii) Permease

(c) a gene (iii) Repressor

(d) y gene (iv) Transacetylase

Select the correct option.

(a) (b) (c) (d)

(1) (i) (iii) (ii) (iv)

(2) (iii) (i) (ii) (iv)

(3) (iii) (i) (iv) (ii)

(4) (iii) (iv) (i) (ii)

64. What will be the sequence of mRNA produced by the following stretch of DNA ?

3'ATGCATGCATGCATG5' - TEMPLATE STRAND **[NEET-2019(Odisha)]**

5'TACGTACGTACGTAC3' - CODING STRAND

(1) 3'AUGCAUGCAUGCAUG5' (2) 5'UACGUACGUACGUAC3'

(3) 3'UACGUACGUACGUAC5' (4) 5'AUGCAUGCAUGCAUG3'

65. Match the following RNA polymerase with their transcribed products :- **[NEET-2019(Odisha)]**

(a) RNA polymerase I (i) tRNA

(b) RNA polymerase II (ii) rRNA

(c) RNA polymerase III (iii) hnRNA

(1) a-i, b-iii, c-ii

(2) a-i, b-ii, c-iii

(3) a-ii, b-iii, c-i

(4) a-iii, b-ii, c-i

66. From the following, identify the correct combination of salient features of Genetic Code :-

(1) Universal, Non-ambiguous, Overlapping **[NEET-2019(Odisha)]**

(2) Degenerate, Overlapping, Commaless

(3) Universal, Ambiguous, Degenerate

(4) Degenerate, Non-overlapping, Non ambiguous

67. Which scientist experimentally proved that DNA is the sole genetic material in bacteriophage ?

(1) Beadle and Tautum (2) Messelson and Stahl **[NEET-2019(Odisha)]**

(3) Hershey and Chase (4) Jacob and Monod

68. In the process of transcription in Eukaryotes, the RNA polymerase I transcribes :-

(1) mRNA with additional processing, capping and tailing **[NEET-2019(Odisha)]**

(2) tRNA, 5 S rRNA and snRNAs

(3) rRNAs–28 S, 18 S and 5.8 S

(4) Precursor of mRNA, hnRNA

69. What initiation and termination factors are involved in transcription in prokaryotes ?

(1) σ and ρ, respectively (2) α and β, respectively **[NEET-2019(Odisha)]**

(3) β and γ, respectively (4) α and σ, respectively

70. The experimental proof for semiconservative replication of DNA was first shown in a:-

(1) Fungus (2) Bacterium **[NEET-2018]**

(3) Plant (4) Virus

71. Select the correct match :- **[NEET-2018]**

(1) Alec Jeffreys – Streptococcus pneumoniae

(2) Alfred Hershey and – TMV Martha Chase

(3) Matthew Meselson – Pisum sativum and F. Stahl

(4) Francois Jacob and – Lac operon Jacques Monod

72. Select the correct Match :- **[NEET-2018]**

(1) Ribozyme - Nucleic acid

(2) F2 × Recessive parent - Dihybrid cross

(3) T.H. Morgan - Transduction

(4) G. Mendel - Transformation

73. AGGTATCGCAT is a sequence from the coding strand of a gene. What will be the corresponding sequence of the transcribed mRNA ? **[NEET-2018]**

(1) AGGUAUCGCAU (2) UGGTUTCGCAT

(3) ACCUAUGCGAU (4) UCCAUAGCGUA

74. All of the following are part of an operon except :- **[NEET-2018]**

(1) an operator (2) structural genes

(3) an enhancer (4) a promoter

75. The final proof for DNA as the genetic material came from the experiments of :- **[NEET-2017]**

(1) Hershey and Chase (2) Avery, Mcleod and McCarty

(3) Hargobind Khorana (4) Griffith

76. DNA fragments are:- **[NEET-2017]**

(1) Negatively charged

(2) Neutral

(3) Either positively or negatively charged depending on their size

(4) Positively charged

77. If there are 999 bases in an RNA that codes for a protein with 333 amino acids, and the base at position 901 is deleted such that the length of the RNA becomes 998 bases, how many codons will be altered? **[NEET-2017]**

(1) 11 (2) 33

(3) 333 (4) 1

78. During DNA replication, Okazaki fragments are used to elongate:- **[NEET-2017]**

(1) The lagging strand towards replication fork.

(2) The leading strand away from replication fork.

(3) The lagging strand away from the replication fork.

(4) The leading strand towards replication fork.

79. Which of the following RNAs should be most abundant in animal cell ? **[NEET-2017]**

(1) t-RNA (2) m-RNA

(3) mi-RNA (4) r-RNA

80. What is the criterion for DNA fragments movement on agarose gel during gel electrophoresis ?

(1) The smaller the fragment size, the farther it moves **[NEET-2017]**

(2) Positively charged fragments move to farther end

(3) Negatively charged fragments do not move

(4) The larger the fragment size, the farther it moves

81. DNA replication in bacteria occurs:- **[NEET-2017]**

(1) Within nucleolus (2) Prior to fission

(3) Just before transcription (4) During S phase

82. Spliceosomes are not found in cells of:- **[NEET-2017]**

(1) Fungi (2) Animals

(3) Bacteria (4) Plants

83. The association of histone H1 with a nucleosome indicates:- **[NEET-2017]**

(1) DNA replication is occurring.

(2) The DNA is condensed into a Chromatin Fibre.

(3) The DNA double helix is exposed.

(4) Transcription is occurring.

84. Which of the following is required as inducer(s) for the expression of Lac operon? **[NEET- I 2016]**

(1) Glucose (2) Galactose

(3) Lactose (4) Lactose and galactose

85. A complex of ribosomes attached to a single strand of RNA is known as :- **[NEET- I 2016]**

(1) Polysome (2) Polymer

(3) Polypeptide (4) Okazaki fragment

86. Which of the following is not required for any of the techniques of DNA fingerprinting available at present? **[NEET- I 2016]**

(1) Polymerase chain reaction (2) Zinc finger analysis

(3) Restriction enzymes (4) DNA–DNA hybridization

87. Which one of the following is the starter codon? **[NEET- I 2016]**

(1) AUG (2) UGA

(3) UAA (4) UAG

88. Taylor conducted the experiment to prove semiconservative mode of chromosome replication on

(1) Drosophila melanogaster (2) E. coli **[NEET- II 2016]**

(3) Vinca rosea (4) Vicia faba

89. The equivalent of a structural gene is :- **[NEET- II 2016]**

(1) Operon (2) Recon

(3) Muton (4) Cistron

90. Which of the following rRNAs acts as structural RNA as well as ribozyme in bacteria?

(1) 23 S rRNA (2) 5.8 S rRNA **[NEET- II 2016]**

(3) 5 S rRNA (4) 18 S rRNA

91. A non-proteinaceous enzyme is :- **[NEET- II 2016]**

(1) Ligase (2) Deoxyribonuclease

(3) Lysozyme (4) Ribozyme

92. A molecule that can act as a genetic material must fulfill the traits given below, except :-

(1) It should be unstable structurally and chemically **[NEET- II 2016]**

(2) It should provide the scope for slow changes that are required for evolution

(3) It should be able to express itself in the form of 'Mendelian characters'

(4) It should be able to generate its replica

93. DNA-dependent RNA polymerase catalyzes transcription on one strand of the DNA which is called the :- **[NEET- II 2016]**

(1) Alpha strand (2) Antistrand

(3) Template strand (4) Coding strand

94. The mechanism that causes a gene to move from one linkage group to another is called:-

(1) Translocation (2) Crossing-over **[NEET- II 2016]**

(3) Inversion (4) Duplication

ANSWER KEY

1. (3)
2. (3)
3. (2)
4. (1)
5. (4)
6. (1)
7. (3)
8. (4)
9. (1)
10. (3)
11. (4)
12. (1)
13. (4)
14. (3)
15. (2)
16. (4)
17. (1)
18. (4)
19. (1)
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89. (4)
90. (1)
91. (4)
92. (1)
93. (3)
94. (1)